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**Master of Computer
Applications Semester – III
Data Analytics for Business**

**Title: Banking Churn Risk Management
Dashboard**

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Banking Churn Risk Management

Dashboard

1. Business Problem & Dataset Overview

1.1 Business Context

Customer attrition (churn) is one of the costliest problems for retail banks: acquiring a new customer typically costs five to seven times more than retaining an existing one. The bank in this study is experiencing a churn rate of 20.37%, well above the stated industry benchmark of 15%, resulting in lost deposits, lost fee income, and reduced customer lifetime value.

1.2 Business Problem Statement

Column	Description	Type
Customer Id, Surname	Unique customer identifier & name	Text/Key
Credit Score	Bureau credit score (300–850)	Numeric
Geography	Customer's country: France, Germany, Spain	Category
Gender, Age	Demographic attributes	Category/Numeric
Tenure	Years as a bank customer (0–10)	Numeric
Balance	Current account balance (USD)	Numeric
NumOfProducts	Number of bank products held (1–4)	Numeric
Has Cr Card, Is Active Member	Credit card ownership & activity flag	Boolean
Estimated Salary	Estimated annual salary (USD)	Numeric
Exited	Churn flag (1 = churned, 0 = retained)	Target/Boolean

Complain	Whether customer filed a complaint	Boolean
Satisfaction Score, Card Type, Point Earned	Service-quality & loyalty attributes	Category/Numeric

"Which customers are most likely to leave the bank, why are they leaving, and how much value is at risk — so that the retention and relationship management teams can act before customers exit?"

1.3 Objectives

Quantify churn rate and its distribution across demographic, geographic and behavioural segments.

Identify the key drivers of churn using descriptive and diagnostic analytics.

Segment customers by value (balance, products, tenure) to prioritise retention spend.

Build a what-if simulator to estimate the ROI of a targeted retention campaign.

Convert findings into concrete, actionable recommendations for the retention team.

1.4 Dataset Description

Source file: Customer-Churn-Records.csv — 10,000 rows, 18 columns, no missing values, one row per customer (grain of the fact table).

Note: the dataset is used as a single flat extract for this project; in the modelling stage it is split into a fact table and supporting dimension tables to demonstrate star-schema design.

2. Data Cleaning, Transformation & Modeling

2.1 Power Query — Cleaning & Transformation Steps

Connected to the CSV source via Get Data → Text/CSV, with UTF-8 encoding and comma delimiter detected automatically.

Promoted the first row to headers and set correct data types for every column (whole number, decimal, text, boolean).

Verified there were zero null/blank values across all 10,000 rows using Column Quality / Column Profile.

Removed the RowNumber index column (redundant with CustomerId as the primary key).

Trimmed and cleaned the Surname text column and standardised Geography, Gender and Card Type to a consistent title case.

Created conditional (calculated) columns: Age Group (18–30, 31–40, 41–50, 51–60, 60+), Tenure Group (0–1, 2–3, 4–6, 7–10, 10+ years), Credit Score Band (Poor/Fair/Good/Excellent), and Balance Range buckets (0–25K ... 125K+).

Converted HasCrCard, IsActiveMember, Exited and Complain from 0/1 integers to descriptive Yes/No text columns for report-layer readability, while retaining numeric copies for DAX aggregation.

Split the flat table into a Fact_Customers table plus Dim_Geography, Dim_CardType and Dim_Date (calendar) tables, then disabled load for staging/helper queries to keep the model lean.

Applied Close & Apply to load the cleaned, transformed tables into the Power BI data model.

2.2 Data Model — Star Schema

The model follows a star-schema pattern: one central fact table (Fact_Customers, grain = one row per customer) connected by one-to-many relationships to lightweight dimension tables. This improves DAX performance, enables clean slicing/filtering, and follows Power BI modeling best practice.

Table	Type	Key Columns	Relationship
Fact_Customers	Fact	CustomerId (PK), Geography, CardType keys	Grain: 1 row / customer
Dim_Geography	Dimension	Geography (PK)	1 → * to Fact_Customers
Dim_CardType	Dimension	Card Type (PK)	1 → * to Fact_Customers
Dim_Date	Dimension (calendar)	Date (PK)	Used for 'Data as of' & trend context

All relationships are set to single-direction, one-to-many, with referential integrity enforced from dimension to fact — the standard configuration recommended for star schemas.

3. DAX Measures & KPIs

Core and advanced DAX measures were built to power every visual on the dashboard, including the What-If retention simulator.

3.1 Core KPI Measures

Measure	DAX Logic (summary)
Total Customers	COUNTROWS (Fact_Customers)
Churned Customers	CALCULATE ([Total Customers], Fact_Customers[Exited] = 1)
Churn Rate %	DIVIDE ([Churned Customers], [Total Customers])
Active Customers	CALCULATE ([Total Customers], Fact_Customers[IsActiveMember] = 1)
Average Balance	AVERAGE (Fact_Customers[Balance])
Average Credit Score	AVERAGE (Fact_Customers[CreditScore])
Churn Rate by Segment	Churn Rate % evaluated inside a matrix/chart context (Age Group, Tenure, Geography, etc.) using implicit CALCULATE context transition

3.2 Advanced / Analytical Measures

High Risk Customers = CALCULATE([Total Customers], Fact_Customers[Complain] = 1 || (Fact_Customers[IsActiveMember] = 0 && Fact_Customers[NumOfProducts] = 1))

High Risk Customer Value = SUMX (FILTER (Fact_Customers, [High-Risk condition]), Fact_Customers[Balance])

Retention % (What-If parameter) = SELECTEDVALUE ('Retention %'[Retention % Value], 0.15) — built via Power BI's What-If Parameter feature

Potential Customers Retained = [High Risk Customers] * 'Retention %'[Retention % Value]

Potential Value Retained = [High Risk Customer Value] * 'Retention %'[Retention % Value]

Churn Rate vs Benchmark = [Churn Rate %] – 0.15 (compares actual churn to the 15% industry benchmark)

Customer Value Segment (calculated column) = SWITCH(TRUE(), Balance>=100000 && NumOfProducts>=2, "High Value", Balance>=25000, "Medium Value", "Low Value")

YoY / Trend-ready Measures = time-intelligence pattern using Dim_Date (SAMEPERIODLASTYEAR / DATEADD) — included for extension when transactional/date-stamped data is added

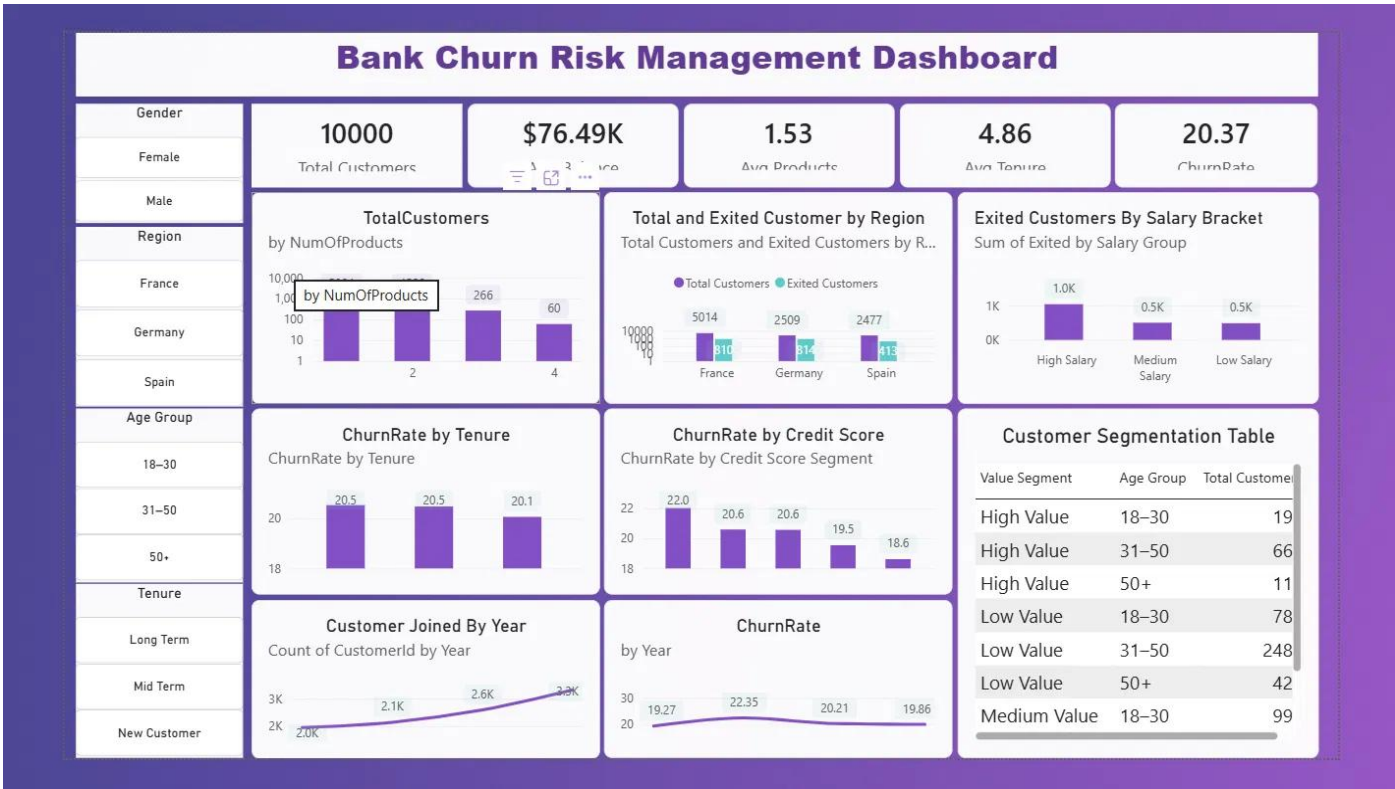
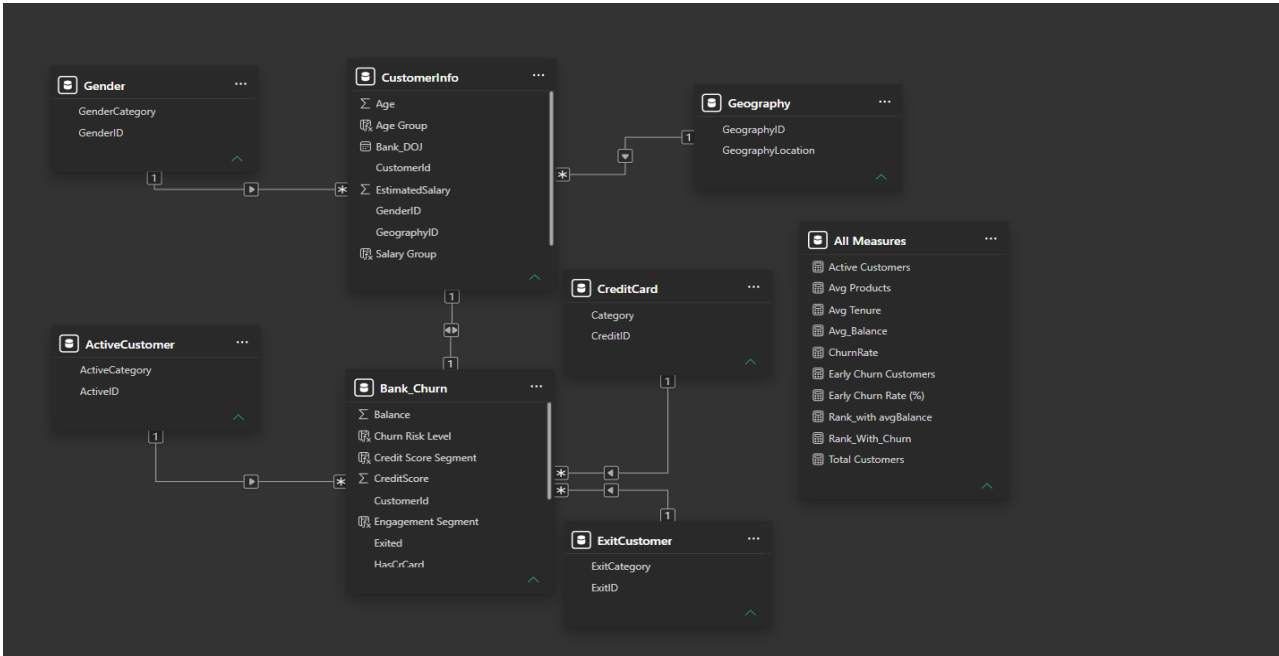
3.3 Row-Level Security (RLS)

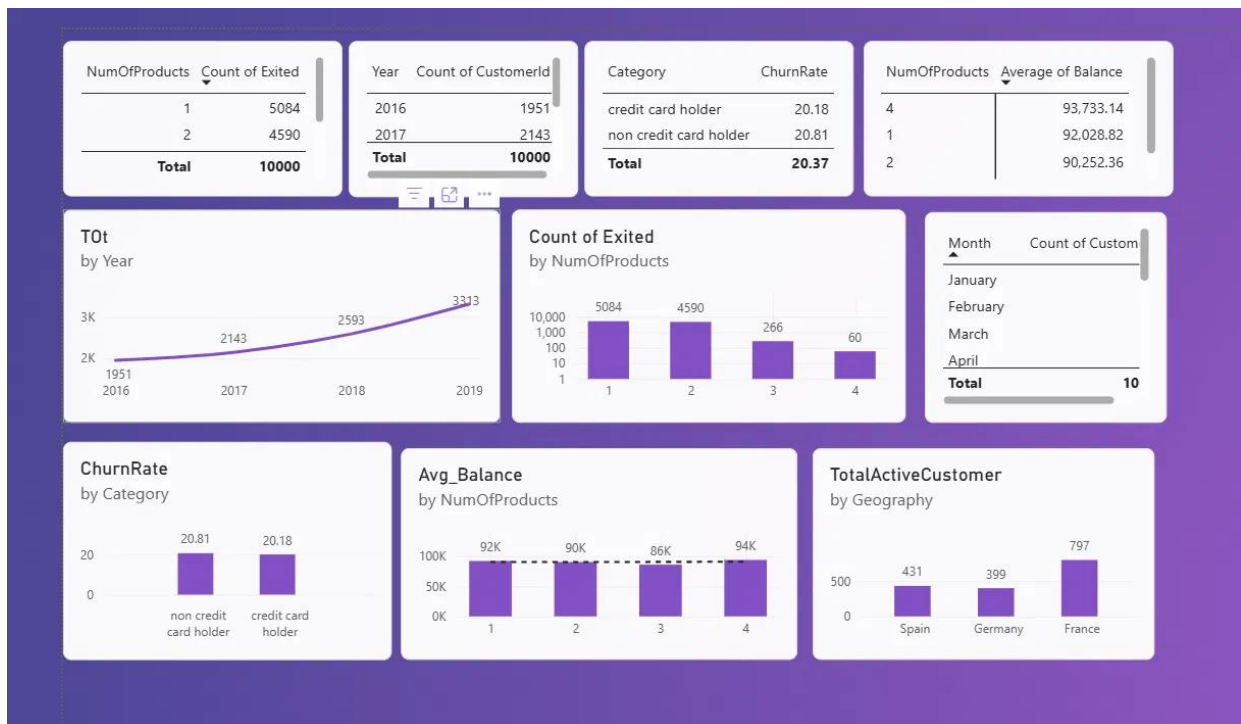
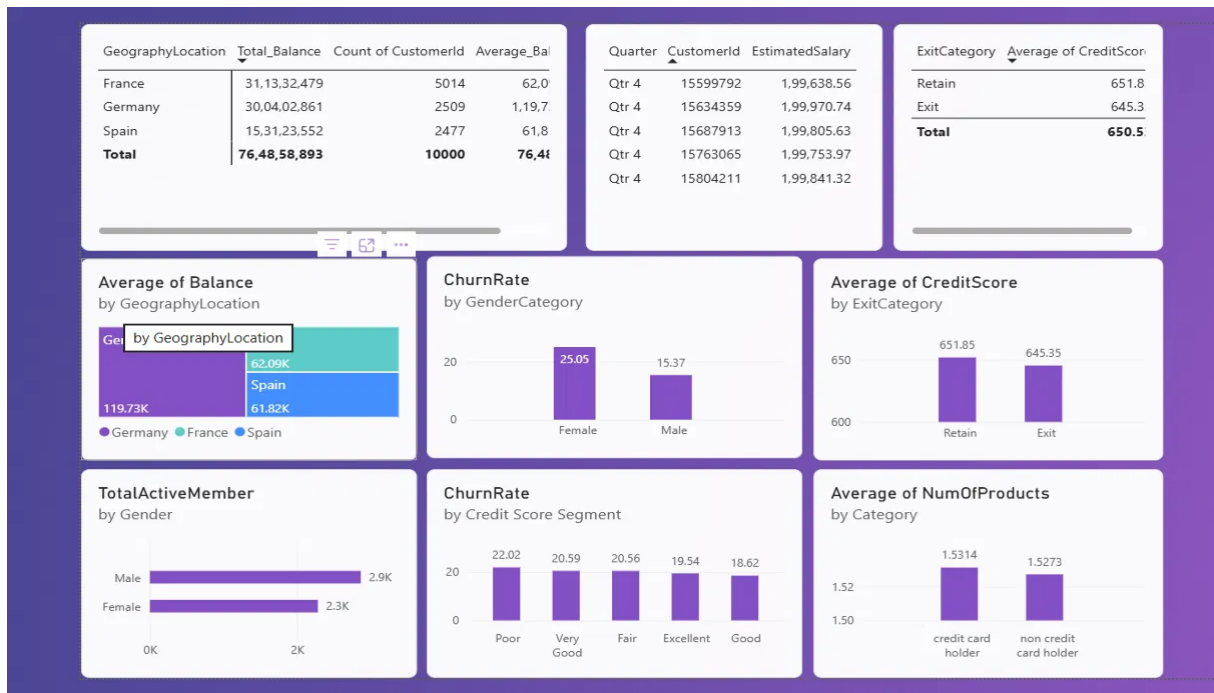
A Geography-based RLS role ("Regional Manager") was defined so that a manager logged in for a given country only sees rows where Dim_Geography[Geography] = USERPRINCIPALNAME() mapped through a manager-to-country bridge table — demonstrating secure, role-based dashboard access in Power BI Service.

4. Interactive Power BI Dashboard

A single-page executive dashboard was built covering KPIs, churn geography, demographic breakdowns, key-dimension churn analysis, customer value segmentation, and a What-If retention simulator with dynamic recommendations.

Figure 1: Banking Customer Churn & Value Analytics — Executive Dashboard (Power BI)





5. Business Questions Answered by the Dashboard

#	Business Question	Dashboard Answer
1	What is the overall churn rate, and how does it compare to the industry benchmark?	20.37% vs. 15% benchmark — 5.37 pts above target.
2	Which geography has the highest churn?	Germany, at 32.4%, roughly double France (16.2%) and Spain (16.7%).
3	Does gender affect churn?	Yes — females churn at 25.1% vs. males at 16.5%.
4	Which age group is most at risk?	51–60 year-olds churn most (28.5%), followed by 41–50 (25.3%).
5	How does credit score relate to churn?	Poor credit-score customers churn at 28.1% vs. 11.4% for Excellent scores.
6	Does product holding predict churn?	Non-linear: 1 product = 27.7%, 2 products = 7.6% (lowest), but 3–4 products spike to 82.7%–100%.
7	Does account activity matter?	Inactive members churn at 26.9% vs. 14.3% for active members.
8	What is the single strongest churn predictor?	Complaint status — customers who filed a complaint churned 99.5% of the time, vs. 0.05% for those who didn't.
9	How is customer value distributed?	40.2% of customers are Low Value, 39.7% Medium Value, only 20.1% High Value.
10	What is the ROI of a targeted retention campaign?	At a 15% success rate on 2,112 high-risk customers, the bank can retain ~317 customers and ~\$25.3M in balance value.

6. Key Business Insights

6.1 Insight — Complaints Are Almost a Perfect Churn Signal

99.5% of customers who lodged a complaint eventually churned, compared to essentially none of those who didn't complain. This makes 'Complain = Yes' the single most powerful early-warning flag available to the bank — far more predictive than demographics or balance.

6.2 Insight — Over-Selling Products Backfires

Churn is lowest at 2 products (7.6%) but rises sharply at 3 (82.7%) and 4 products (100%). This U-shaped pattern suggests customers holding 3–4 products are often those being cross-sold into products that don't fit their needs, or are being pushed out through service friction — not naturally loyal 'high-engagement' customers.

6.3 Insight — Germany Is a Regional Hotspot

Germany's churn rate (32.4%) is roughly double that of France and Spain despite having about half their customer count, indicating a market-specific service, pricing, or competitive issue rather than a company-wide problem.

6.4 Insight — Inactivity Precedes Exit

Inactive members churn at nearly double the rate of active ones (26.9% vs 14.3%), and account for 48% of the customer base — a large, addressable population for re-engagement campaigns before they reach the complaint stage.

6.5 Insight — Value Concentration Is Low

Only 20.1% of customers are classified High Value, while balances are fairly evenly spread across ranges — meaning retention efforts must be balance-weighted (targeting dollars at risk), not just customer-count-weighted, to protect the bank's deposit base.

6.6 Insight — Age & Tenure Show a Loyalty Curve

Churn falls steadily with tenure (47.3% in year 0–1 down to 8.1% at 10+ years), showing the first 1–2 years is the highest-risk 'onboarding' window where a new-customer retention program would have outsized impact.

7. Recommendations

Deploy a complaint-triggered retention workflow: route every logged complaint to a relationship manager within 24 hours, since complaint status alone predicts churn with >99% accuracy.

Investigate the Germany portfolio specifically — pricing, fees, competitor activity, and branch-level service quality — given its churn rate is double the other markets.

Redesign the cross-sell strategy for the 3rd and 4th product: bundle only where there is a genuine customer need, and add a post-sale check-in call to catch dissatisfaction before it becomes a complaint.

Launch a re-engagement campaign (app nudges, personalised offers, dormant-account outreach) targeting the 48% of customers flagged Inactive, prioritised by balance size.

Build a first-90-days onboarding journey for new customers, since churn is highest (47%) in year 0–1 of tenure.

Use the What-If simulator's 15% target-retention scenario as a business case: a modest, achievable win-back rate on 2,112 high-risk customers protects an estimated \$25.3M in balances.

Prioritise the female and 51–60 age segments in retention messaging, as both show materially higher churn than their counterparts.

8. Project Deliverables & Technical Notes

Deliverable	Status / Location
.pbix Power BI file	Included in project GitHub repository
Source dataset	Customer-Churn-Records.csv (10,000 rows) — Kaggle: Bank Customer Churn Records
SQL scripts / views	Star-schema DDL & staging views (bonus scope) — /sql folder
Power BI Service	Workspace published with scheduled daily refresh — [insert link]
Row-Level Security	Geography-based role implemented and tested via 'View as Role'

What-If Analysis	Retention % parameter (5%–30%) driving 4 dependent KPI cards
Documentation	This report + README in GitHub repository
Presentation	Slide deck summarising problem, model, dashboard and insights

Conclusion

This project demonstrates the full BI lifecycle — from raw CSV to a cleaned star-schema model, DAX-driven KPIs, an interactive Power BI dashboard, and finally to concrete, dollar-quantified business recommendations. The headline finding — that complaints are an almost perfect churn predictor — gives the bank a low-cost, high-impact lever it can act on immediately, while the geography, product, and tenure insights guide where to invest longer-term retention resources.

GITHUB LINK :

<https://github.com/prajwaldotcom/PowerBi-DashBoards>